

VideoIndex benchmark report

Retrieval and question answering over 36.6 hours of lectures and workshops, indexing throughput, and a comparison with Gemini agentic video

What was measured, how, and what the numbers mean: the 30-video development dataset, the two dev sets and their scoring rules, retrieval quality before and after the full-text query fix, question answering with the agent and a retrieval-only baseline, indexing throughput on CPU and GPU, and a like-for-like run of Gemini's agentic video understanding on the same questions and files.

Generated from the evaluation artefacts by `scripts/report/benchmark_report.py`
2026-09-12 · c38b5a6

Contents

VideoIndex benchmark report	3
Summary	3
Dataset	3
Methodology	5
Results: retrieval	7
Results: question answering	8
Results: fine index against coarse index	11
Comparison with Gemini agentic video	11
Results: indexing throughput	15
Reproducing	19

VideoIndex benchmark report

Summary

Retrieval hit@5

0.94

72 questions, k = 5, 30 s tolerance

Retrieval MRR

0.72

was 0.50 before the query fix

QA accuracy, agent

96.2%

52 questions, 100% cited, \$0.052 each

QA accuracy, retrieval-only

73.1%

one search then answer, \$0.017 each

This report documents the measurements made on the VideoIndex development dataset on 2026-09-11 and 2026-09-12. It is generated from the evaluation artefacts (`/data/videoindex/eval/*.json`), the indexing logs and the index itself, so every number can be regenerated by re-running the scripts named in each section. The dataset is 30 lecture and workshop recordings totalling 36:34:35 of video; the index built from it holds 1.98 GiB.

Three results stand out.

- 1. Retrieval quality was gated by one bug, not by the models.** The full-text query required every word of a question, so natural-language questions got no BM25 candidates at all and the fusion ran on the vector lists alone. OR-ing the terms took hit@5 from 0.77 to 0.94 and MRR from 0.50 to 0.72.
- 2. The agent earns its cost.** Against a retrieval-only baseline that answers from one search (73.1%), the tool-using agent reaches 96.2% at 3.0× the cost per question (\$0.052 against \$0.017) and about 2.0 tool calls per question.
- 3. Indexing is OCR-bound, not decode-bound, and the GPU fixes that** once cuDNN's exhaustive per-shape algorithm search is turned off: the visual pass on a 47-minute lecture runs in 50 s on the CUDA build against 112 s on the CPU build, with decode alone at 39.5 s.
- 4. Gemini 3.8 Flash agentic video, asked 20 of the same questions over the same files, scored 95.0% at \$0.047 and 5.9 s median per question** (VideoIndex agent on those 20: 100.0%, \$0.049, 6.5 s). Both systems answer these developer-written questions well; the discriminating comparison waits for the public long-video benchmarks.

Dataset

The development dataset is the two YouTube playlists in `dataset/videolist.md`: the AI Engineer conference workshop playlist and the UC Berkeley Agentic AI MOOC (CS294-196, Fall 2025). They were chosen because they are long, lecture-style and slide-heavy, which stresses the parts of the system that matter for hour-scale video: transcript search over an hour of speech, on-screen text, chapter segmentation and the agent's ability to find a moment rather than summarise a clip. The videos were downloaded at 720p with subtitles, chapters and `.info.json` metadata, and indexed with the `coarse_only` policy (subtitle import, VAD, Whisper large-v3, 1 fps sampling, pHash, thumbnails, shot boundaries, SigLIP frame embeddings, RapidOCR on changed frames, bge-small text embeddings).

Video	Channel	Duration	Transcript spans	OCR lines	Frames (1 fps)	Shots
[Full Workshop] Reinforcement Learning, Kernels, Reasoning, Quantizati	AI Engineer	2:42:27	699	22,663	9,748	155
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / LLM Agents Overvie	Berkeley RDI	1:58:21	1,068	2,148	7,101	151

Video	Channel	Duration	Transcript spans	OCR lines	Frames (1 fps)	Shots
Building Code First AI Agents with Azure AI Agent Service — Cedric Vid	AI Engineer	1:54:05	493	11,034	6,846	169
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Agentic AI Safety	Berkeley RDI	1:48:50	1,032	3,602	6,530	101
RFT, DPO, SFT: Fine-tuning with OpenAI — Ilan Bigio, OpenAI	AI Engineer	1:46:14	459	35,289	6,375	136
How to build world-class AI products — Sarah Sachs (AI lead @ Notion)	AI Engineer	1:43:45	458	10,200	6,226	123
How LLMs work for Web Devs: GPT in 600 lines of Vanilla JS - Ishan Ana	AI Engineer	1:41:33	428	8,070	6,094	146
Shipping AI That Works: An Evaluation Framework for PMs – Aman Khan, A	AI Engineer	1:26:16	377	6,275	5,176	125
VoiceVision RAG - Integrating Visual Document Intelligence with Voice	AI Engineer	1:23:51	371	10,024	5,032	113
A2A & MCP Workshop: Automating Business Processes with LLMs — Damien M	AI Engineer	1:23:13	368	1,451	4,994	108
[Full Workshop] Vibe Coding at Scale: Customizing AI Assistants for En	AI Engineer	1:20:38	353	23,800	4,838	109
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Evolution of Syste	Berkeley RDI	1:19:31	698	1,021	4,772	81
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Post-Training Veri	Berkeley RDI	1:17:28	688	1,117	4,649	60
Full Workshop: Realtime Voice AI — Mark Backman, Daily	AI Engineer	1:09:40	295	5,963	4,181	88
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Training Agentic M	Berkeley RDI	1:04:37	565	742	3,877	61
Collaborating with Agents in your Software Dev Workflow - Jon Peck & C	AI Engineer	1:04:06	265	6,221	3,846	81
[Full Workshop] Building Conversational AI Agents - Thor Schaeff, Elev	AI Engineer	1:01:42	275	3,349	3,702	86
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Autonomous Agents	Berkeley RDI	1:01:42	550	3,443	3,702	134
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / AI Agents to Autom	Berkeley RDI	1:01:15	544	1,401	3,676	62
Agentic AI MOOC / UC Berkeley CS294-196 F25 / Multi-Agent Systems in E	Berkeley RDI	0:58:58	527	883	3,539	82
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Multi-Agent AI by	Berkeley RDI	0:54:49	489	675	3,290	49
From Mixture of Experts to Mixture of Agents with Super Fast Inference	AI Engineer	0:53:15	233	4,421	3,195	68
ComfyUI Full Workshop — first workshop from ComfyAnonymous himself!	AI Engineer	0:51:25	232	1,281	3,085	61
Evals 101 — Doug Guthrie, Braintrust	AI Engineer	0:48:31	209	1,453	2,911	76
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Practical Lessons	Berkeley RDI	0:46:54	416	841	2,815	67
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Predictable Noise	Berkeley RDI	0:44:04	409	1,208	2,644	42

Video	Channel	Duration	Transcript spans	OCR lines	Frames (1 fps)	Shots
Introduction to LLM serving with SGLang - Philip Kiely and Yineng Zhan	AI Engineer	0:43:42	194	6,882	2,622	64
Building Multimodal AI Agents From Scratch — Apoorva Joshi, MongoDB	AI Engineer	0:36:58	161	859	2,218	66
The AI Engineer's Guide to Raising VC — Dani Grant (Jam), Chelcie Tayl	AI Engineer	0:34:16	145	580	2,057	26
Strategies for LLM Evals (GuideLLM, Im-eval-harness, OpenAI Evals Work	AI Engineer	0:32:28	143	1,420	1,948	54
30 videos		36:34:35	13,144	178,316	131,689	2,744

Every video is a VP9 or AV1 stream at 1280×720. Transcript spans come from Whisper large-v3 (fp16, CTranslate2) through the local OpenAI-compatible server; when a video shipped human-authored captions the `subtitle_import` operator imported them as a subtitle track as well. OCR lines are RapidOCR (PP-OCRv4 detector, English PP-OCRv3 recogniser) read on frames whose pixels changed by more than the gate; consecutive identical lines are collapsed. Frames are the 1 fps samples; shots are true camera cuts from an HSV-histogram plus edge-change detector, which on slide lectures is a small number.

Methodology

Two dev sets

Two question sets were written by reading each video's caption digest and contact sheets (`scripts/devset_digest.py`), so questions are grounded in what is actually said or shown. Both live under `dataset/` and are versioned with the code.

Retrieval dev set (`dataset/devset.jsonl` , 72 questions over all 30 videos: 65 transcript, 3 OCR, 4 visual). Each question is a paraphrase of a moment, not the caption text, so lexical search alone cannot answer it. Transcript and OCR questions carry an `anchor` phrase; the evaluation resolves the anchor to a time from the video's captions (searching a window of three consecutive cues, since captions break sentences mid-phrase) or from the index's OCR spans, so ground truth follows the media rather than hand-typed seconds. Visual questions carry a `t0 / t1` range read off the contact sheets. Ground truth is the anchor time minus 5 s to plus 30 s.

QA dev set (`dataset/devset_qa.jsonl` , 52 questions over 29 videos). Each question names the speaker or session, asks for a fact with a short answer (a number, a name, a term, a claim) and carries a list of accepted strings and an anchor phrase for the moment that answers it. Accepted strings include common variants (`2,000` , `2000` , `two thousand`).

Retrieval scoring

`scripts/devset_eval.py` runs `vi search` for every question with `k = 5` and hybrid retrieval (BM25 per span kind, bge-small text vectors with the query instruction, SigLIP text-to-frame vectors, reciprocal-rank fusion with `k = 60`, grouping into shots or 60-second pieces). A result is a hit when its video is the question's video and its time range, widened by 30 s, overlaps the ground-truth range. Reported metrics:

Metric	Definition
video@1	the top result is in the right video
hit@1, hit@5	a hit at rank 1; a hit anywhere in the top 5
MRR	mean of $1 / \text{rank}$ of the first hit (0 when none in the top 5)
p50 latency	median wall-clock of the one-shot <code>vi search</code> process, including model load

Question-answering scoring

`scripts/devset_qa_eval.py` runs `vi ask --json` for every question with a budget of \$0.30 and 6 tool calls, `claude-sonnet-5` as the agent model. The answer is **correct** when it contains any accepted string (case-insensitive substring). Citations are the `[[cite:VIDEO:T0-T1]]` markers the agent emits; **cited** means at least one, **citation video correct** means one names the question's video, and **citation in time** means one lies within 90 s of the anchor. Cost is Anthropic list price over every LLM call in the `ask`; tokens are input plus output across those calls; latency is wall-clock per `ask`.

Two agent configurations and one baseline were run:

- **Agent, first run:** the LLM chooses tools (`search`, `get_transcript`, `get_ocr`, `timeline`, `view`, ...) with the original AND-ed full-text query behind `search`.
- **Agent, fixed retrieval:** the same, after OR-ing full-text terms and adding a retry when the model returns an empty final turn after exhausting its tool budget.
- **Retrieval-only baseline:** the `RetrievalOnlyPolicy` issues exactly one `search` with the question (`k = 8`) and the LLM writes the answer from those hits. This is what the index alone gives before any agentic looking.

Known limitations of the scoring

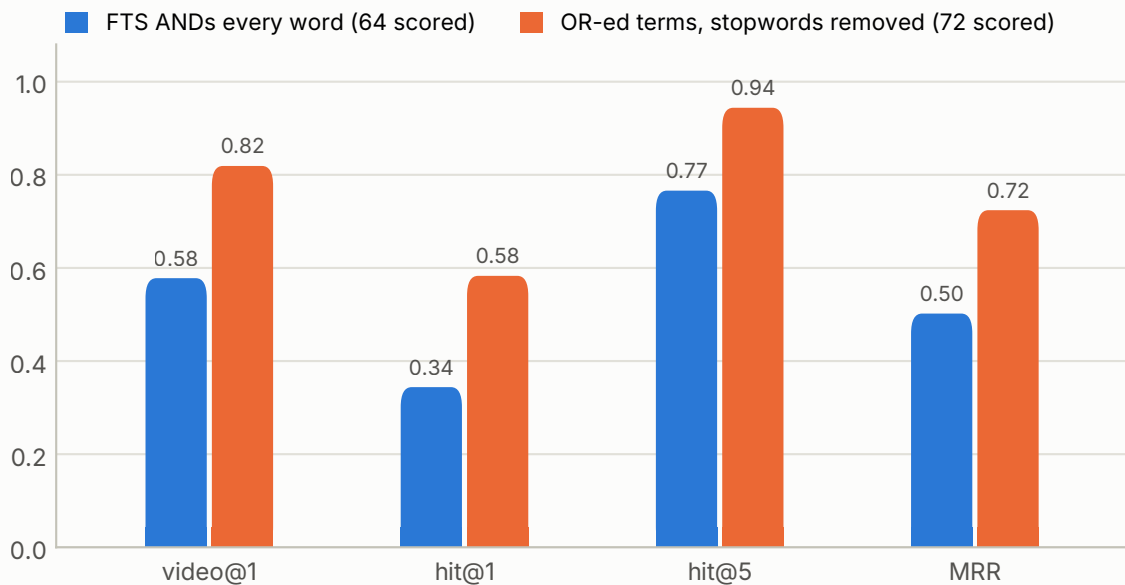
- Substring matching on accepted strings is generous for very short strings (a one-word accept can match inside a longer, wrong answer) and strict for paraphrases (a correct answer phrased differently is a miss). Every system compared below is scored identically, so the comparison is fair even where an individual verdict is debatable; the miss tables show the answers so a reader can judge.
- Anchors mark one moment that answers the question; a speaker who restates the point later produces a "right video, wrong minute" miss in retrieval and a citation-in-time miss in QA.
- The dev sets were written by the developers of the system while it was being built. They are a regression gate and a smoke test for hour-scale retrieval, not a public benchmark; LVBench, Minerva and 1H-VideoQA runs are planned for M4.
- `vi search` latency includes process start and loading the SigLIP text tower and bge-small (about 2 s); the search itself is tens of milliseconds and the Python binding pays the load once.

Hardware and software

All runs used one GCP `a2-ultragpu-1g` (12 vCPUs, 167 GiB RAM, one NVIDIA A100-SXM4-80GB, driver 595.91.07, CUDA 13.1 toolkit, cuDNN 9), Ubuntu 26.04, ffmpeg 8.0.1, rustc 1.98.1, ONNX Runtime 1.28 through `ort` 2.0.0-rc.13, faster-whisper 1.2.1 on CTranslate2 4.8.2. The dataset indexing run used the CPU ONNX build while builds and tests competed for the machine; the GPU timings in the indexing section were taken afterwards on the idle machine.

Results: retrieval

Retrieval on the 72-question dev set, before and after the query fix

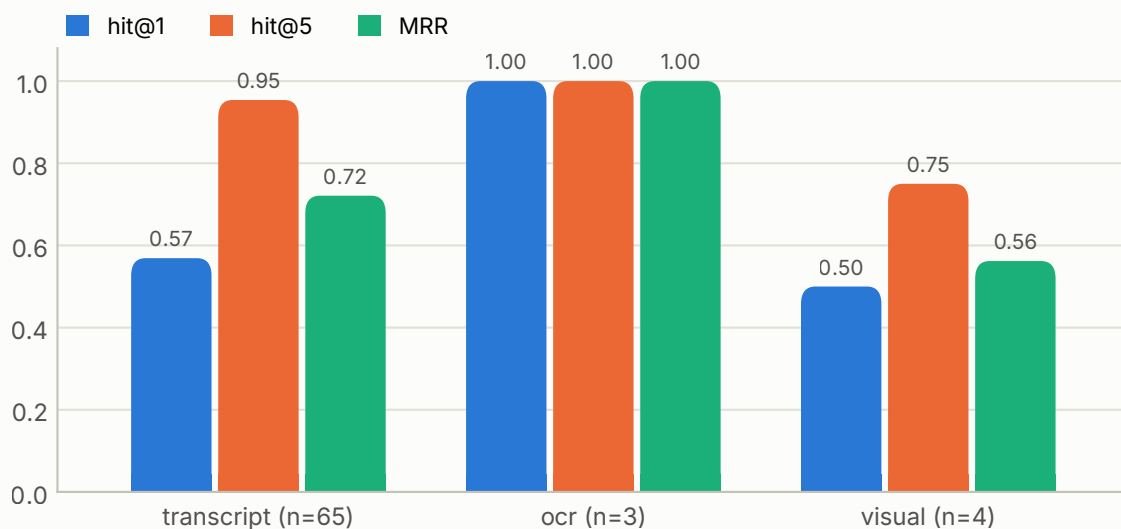


vi search, k = 5, hit = result within 30 s of the anchored ground truth. Source: scripts/devset_eval.py

	n	video@1	hit@1	hit@5	MRR
First run (FTS ANDs every word)	64 scored, 8 anchors unresolved	0.578	0.344	0.766	0.502
After OR-ing terms with stopwords removal	72	0.819	0.583	0.944	0.724

The first run's failure was structural. FTS5 treats a bare list of terms as a conjunction; the question "language models failing to count the Rs in strawberry" became "language" "models" "failing" "to" "count" "the" "Rs" "in" "strawberry"* and matched nothing, so every miss shows `lists=['text_vec', 'image_vec']`: the two BM25 lists were empty and only the two vector lists voted. After the fix the terms are OR-ed (stopwords dropped unless the query is all stopwords, the last term prefix-matched) and BM25 ranks rows that match more terms higher; every hit in the second run carries all four lists.

Retrieval by question type (after the fix)



Question type	n	video@1	hit@1	hit@5	MRR
transcript	65	0.831	0.569	0.954	0.721
OCR (on-screen text)	3	1.000	1.000	1.000	1.000
visual (SigLIP text-to-frame)	4	0.500	0.500	0.750	0.562

The OCR and visual groups are small (3 and 4 questions) and their numbers are indicative only. Visual retrieval uses the SigLIP base patch-16 224 px model on 1 fps frames deduplicated by pHash; a larger SigLIP (so400m, 384 px) is the obvious upgrade when the visual questions grow in number.

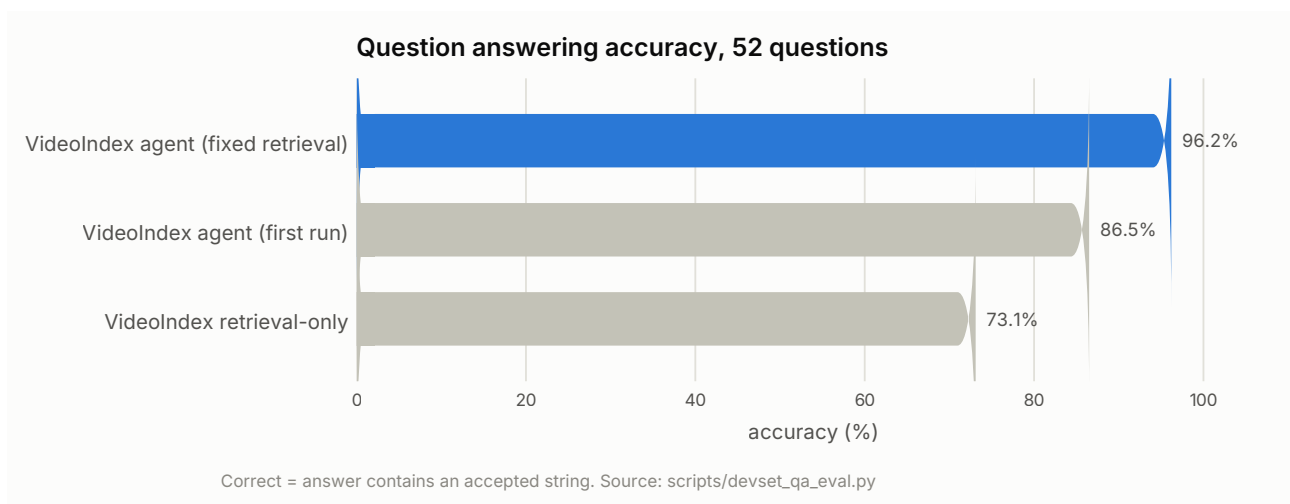
Remaining misses

4 of 72 questions have no hit in the top 5:

Question	Type	What happened
q18: woman in a red shirt holding a phone to her ear	visual	video@4 ; top result 3050–3096 s in ZuiJjkbX0Og
q22: paired versus unpaired standard error when comparing two models on a benchmark	transcript	video@1 ; top result 1862–1922 s in HV8pugcFVO0
q29: grader dependency where one grader depends on another	transcript	video@1 ; top result 1351–1411 s in xNxrBHZPDvM
q32: collision rate as a metric for hardware prompts	transcript	video@None ; top result 2875–2935 s in EAfP8pDs7h4

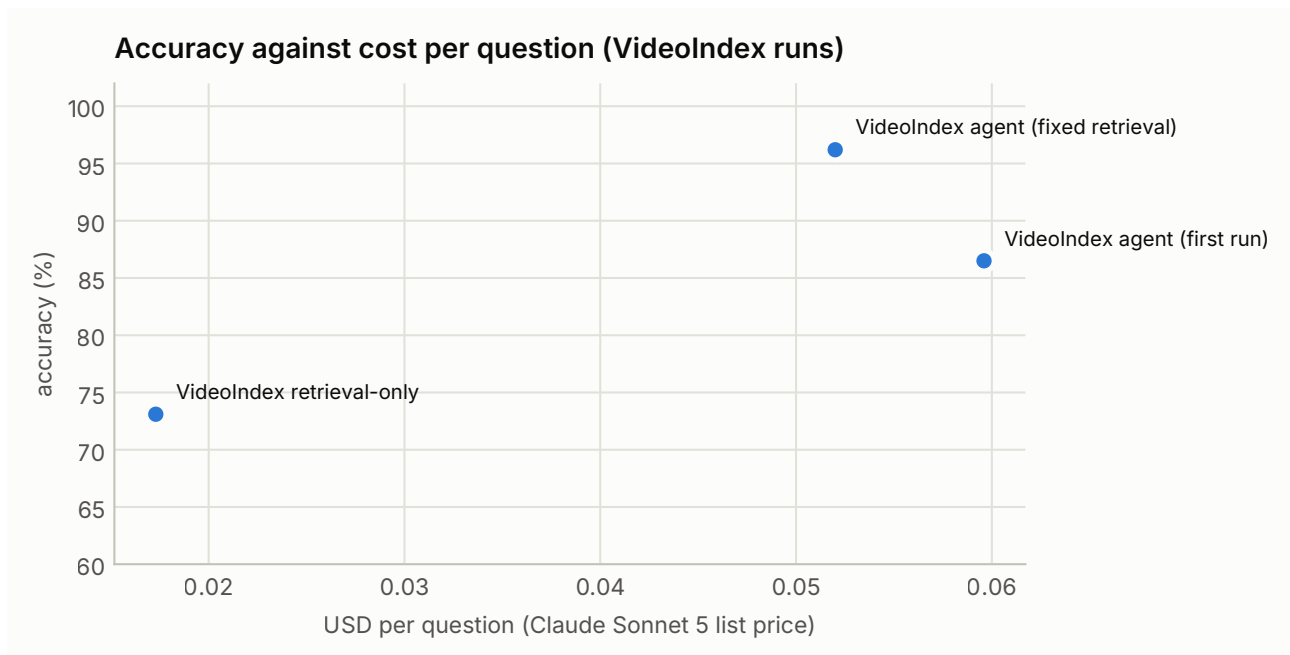
Two are right-video-wrong-minute (the phrasing matches a later restatement), one visual question's frame ranks fourth, and one is a speech-to-text paraphrase gap ("collision rate" is spoken as "collisions") that neither BM25 nor bge-small bridges. Every anchor resolved.

Results: question answering



Run	accuracy	cited	citation video correct	citation within 90 s	cost / question	tokens / question	tool calls / question	median latency	p90 latency
Agent, first run (AND-ed FTS)	86.5%	94.2 %	94.2%	71.2%	\$0.060 (\$3.10 total)	18,053	2.46	7.1 s	12.3 s

Run	accuracy	cited	citation video correct	citation within 90 s	cost / question	tokens / question	tool calls / question	median latency	p90 latency
Agent, fixed retrieval	96.2%	100.0 %	100.0%	80.8%	\$0.052 (\$2.71 total)	15,733	2.04	6.6 s	10.2 s
Retrieval-only baseline	73.1%	75.0 %	75.0%	63.5%	\$0.017 (\$0.90 total)	5,189	1.00	4.6 s	5.4 s



The agent's extra 1.04 tool calls per question buy 23.1 accuracy points. The baseline's misses are mostly questions whose answer sits in a span the first search does not surface; the agent's second search or a `get_transcript` around the first hit finds it. Three of the first run's seven misses were empty answers: the agent spent its six tool calls searching (starved by the AND-ed query) and the final no-tools turn came back with no text. The loop now asks once more, explicitly, before returning a partial answer; with the retrieval fix the agent averages 2.04 calls and never hits the cap.

Agent misses (fixed retrieval)

Question	Accepted	Answer given (truncated)
qa44: What was Notion's first AI feature discussed by Sarah Sachs?	meeting notes	Notion's first AI product, launched around the same time as ChatGPT's release, was called the AI Writer . It let users generate content i
qa52: In Jiantao Jiao's lecture, what metric on the hardware-prompt side is said to be higher wh	collision	In this part of Jiantao Jiao's lecture, he discusses the correlation (covariance) between the model's generation log-probability (confidence

Retrieval-only misses

Question	Accepted	Answer given (truncated)
qa05: In Oriol Vinyals' lecture on AlphaStar, what are the agents called whose only job is to be	exploiter	get_transcript({"video_id":"01M29FVH0TZTRJZ2BEDJWR4TWV","t0":2842,"t1":2900})

Question	Accepted	Answer given (truncated)
qa07: According to Clay Bavor, how many years did the internet take to reach mass adoption, in h	11	get_transcript({"video_id":"01M29B5NW58J2GSKQ2JJY90ESV","t0":200,"t1":320})
qa08: What latency technique does Clay Bavor describe where the first of several parallel reques	speculative request hedging, hedging	Clay Bavor describes using "speculative" parallel inference requests — sending the same request multiple times in parallel and using whi
qa11: In James Zou's lecture, which structure-prediction model do the virtual lab agents use to	alphafold	Ask more specifically about the transcript around 913–964s and the OCR near 834s. get_transcript({"video_id":"01M29G2BQR9RATR7ETSDYGT4C
qa15: In Noam Brown's ultimatum-game example, how much money does Alice offer Bob?	\$30, 30 dollars, thirty	In Noam Brown's ultimatum-game example, Alice is given \$100 and can offer Bob anywhere between \$0 and \$100 . He explains that the theoretica
qa27: In Daniel Han's workshop, how much faster is float4 than float8 according to him?	two times, 2x, twice, 2 times	get_transcript (video_id="01M2920GDE5E53XCVRSC3R8SPA", t0=9590, t1=9660)
qa34: What does Mark Backman ask the voice bot to tell in order to test interruption?	long story, really long story	get_transcript(video_id="01M29J9P5FJCQH56G0CNAR7KJ0", t0=1080, t1=1140)
qa35: Which three tech-forward customers does Aman Khan of Arize name?	uber, reddit, instacart	search({"k":8,"query":"tech-forward customers","video_id":"01M29ABBRDKG9AF3DVQRQBPSN2"})
qa39: Which library does SGLang use for its attention kernels, according to the Baseten workshop	flashinfer, flash infer	query executed: Let me check more specifically for attention kernel libraries mentioned. get_transcript({"video_id":"01M29CBW0MKBB0KNWH
qa40: In Suman Debnath's demo, what are trophic levels?	feeding position, feeding positions, foo	Result of get_transcript({"video_id":"01M29BFEMHGPPDX3PJRNTKG0Q1","t0":4025,"t1":4100})
qa42: In the ElevenLabs workshop, which model identifies the language being spoken?	asr, speech recognition	Let me check the transcript more closely around that segment to find the specific model name. Result of get_transcript({"video_id":"01M29EH
qa44: What was Notion's first AI feature discussed by Sarah Sachs?	meeting notes	Notion's first AI product, discussed by Sarah Sachs, was called the "AI Writer" — it launched around the same time as ChatGPT and let users
qa48: What do the Microsoft presenters ask the audience to do with their phones at the start?	silent, vibrate	search({"query":"phones QR code scan at the beginning", "k":8})

Question	Accepted	Answer given (truncated)
qa52: In Jiantao Jiao's lecture, what metric on the hardware-prompt side is said to be higher wh	collision	Let me look at the transcript around that section in Jiantao Jiao's lecture to find the specific metric mentioned. get_transcript({"video_i

Results: fine index against coarse index

The fine pass (VLM scene descriptions, chapters, entities and events; 5 h 8 min and \$45.96 over the 30 videos with Claude Sonnet 5) was run on the same index and both dev sets re-evaluated.

	Retrieval hit@5	Retrieval MRR	video@1	QA accuracy	citation within anchor	cost / question	tool calls / question
Coarse index	0.944	0.724	0.819	96.2%	80.8%	\$0.052	2.04
Fine index	0.889	0.661	0.708	92.3%	82.7%	\$0.057	2.10

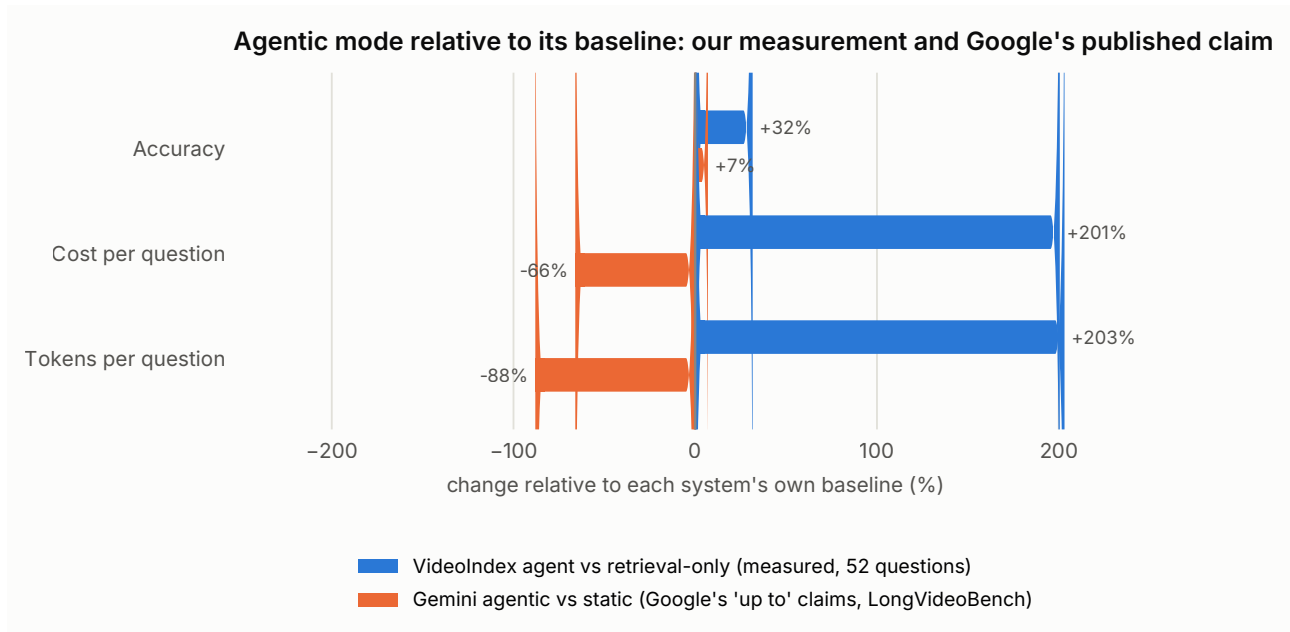
On this transcript-anchored dev set the fine index scores lower: description rows join the fusion as an equal-weight BM25 list and as text vectors, and their plausible neighbours outrank transcript hits under reciprocal-rank fusion (16 retrieval questions moved down, 10 up; two QA questions flipped). Citation-in-anchor improved, so the rows are useful but over-weighted for this question mix. Per-kind list weights are the first tuning target of the M4 sweep; `coarse_only` stays the default for lecture content.

Comparison with Gemini agentic video

What Google published

Google announced agentic video understanding for Gemini 3.7 Flash, 3.6 Flash and 3.5 Flash-Lite on 1 September 2026 (the developer documentation now lists 3.8 Flash as well). With `processing: "agentic"` on a video input, the model "dynamically navigates the video timeline, loading only the content it needs based on the prompt" through internal `processing_call` steps that fetch frames, audio or transcript for a chosen span, instead of the static mode's 1 fps frames at 258 tokens each plus 32 tokens per second of audio (about 300 tokens per second, so a one-hour video is roughly 1.1 million tokens). The claims, quoted from the announcement: agentic mode can "reduce analysis costs by up to 66% and token consumption by up to 88%, while improving accuracy by up to 7%" relative to static processing, with Gemini 3.7 Flash "at the accuracy-to-cost pareto frontier among tested models"; the benchmark named is LongVideoBench, without absolute scores in the post. The feature uses standard token pricing (Gemini 3.x Flash: \$0.75 per million input tokens and \$3.75 per million output tokens through 2026-12-31).

The two systems take different routes to the same goal. Gemini's agent works on the raw video at question time and pays for whatever it loads on every question; VideoIndex indexes once (transcript, on-screen text, embeddings, shots) and answers from the index, decoding pixels only when the agent calls `view`. The published deltas are relative to Gemini's own static mode, so the chart below puts them beside our measured agent-versus-retrieval-only deltas; the direction differs because the baselines differ (a full-video static pass is the expensive end of Gemini's range, while our retrieval-only baseline is the cheap end of ours).



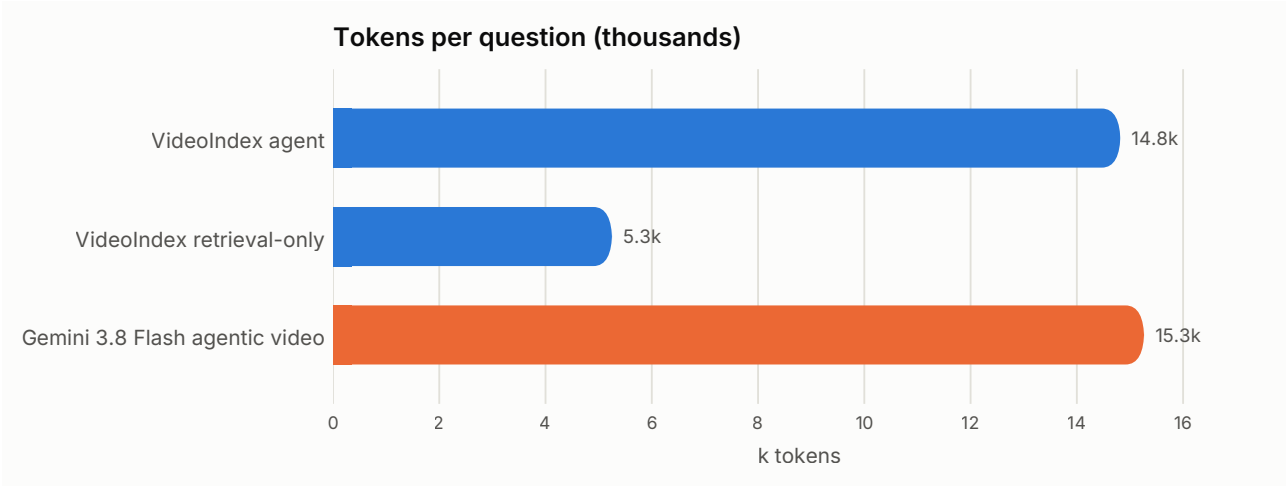
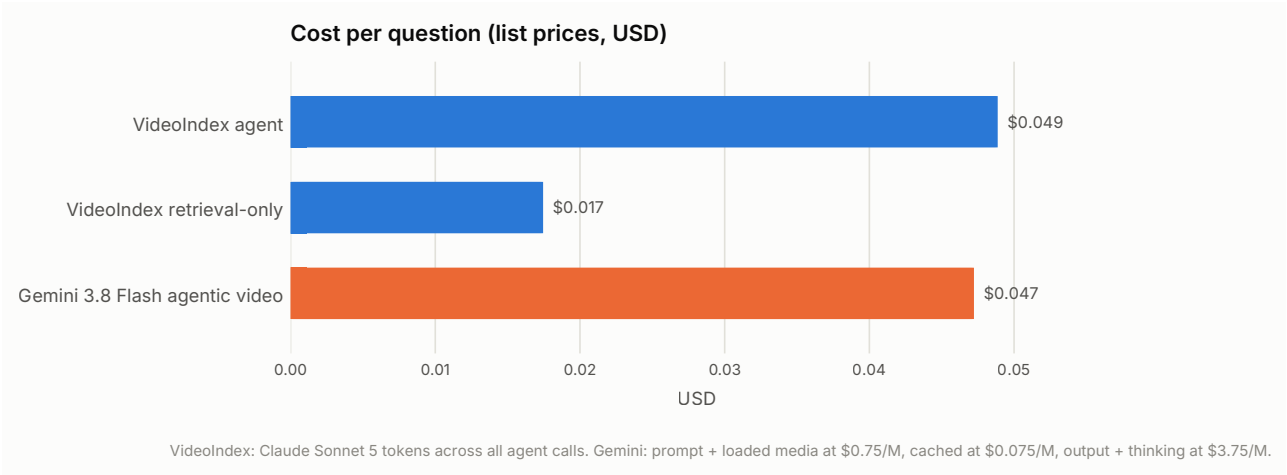
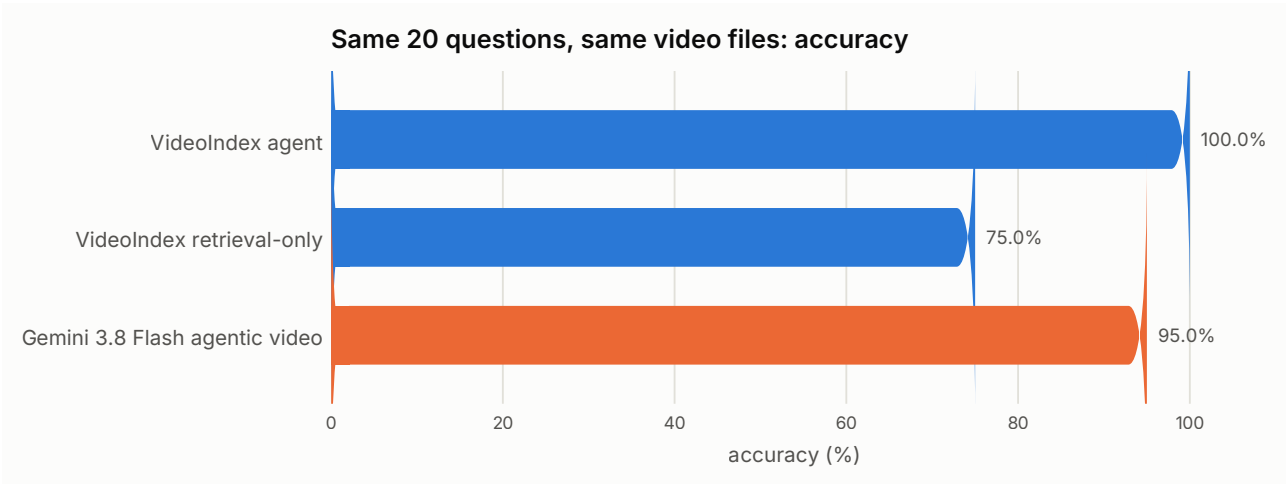
Our own agent-versus-baseline deltas are +32% accuracy, +201% cost and +203% tokens.

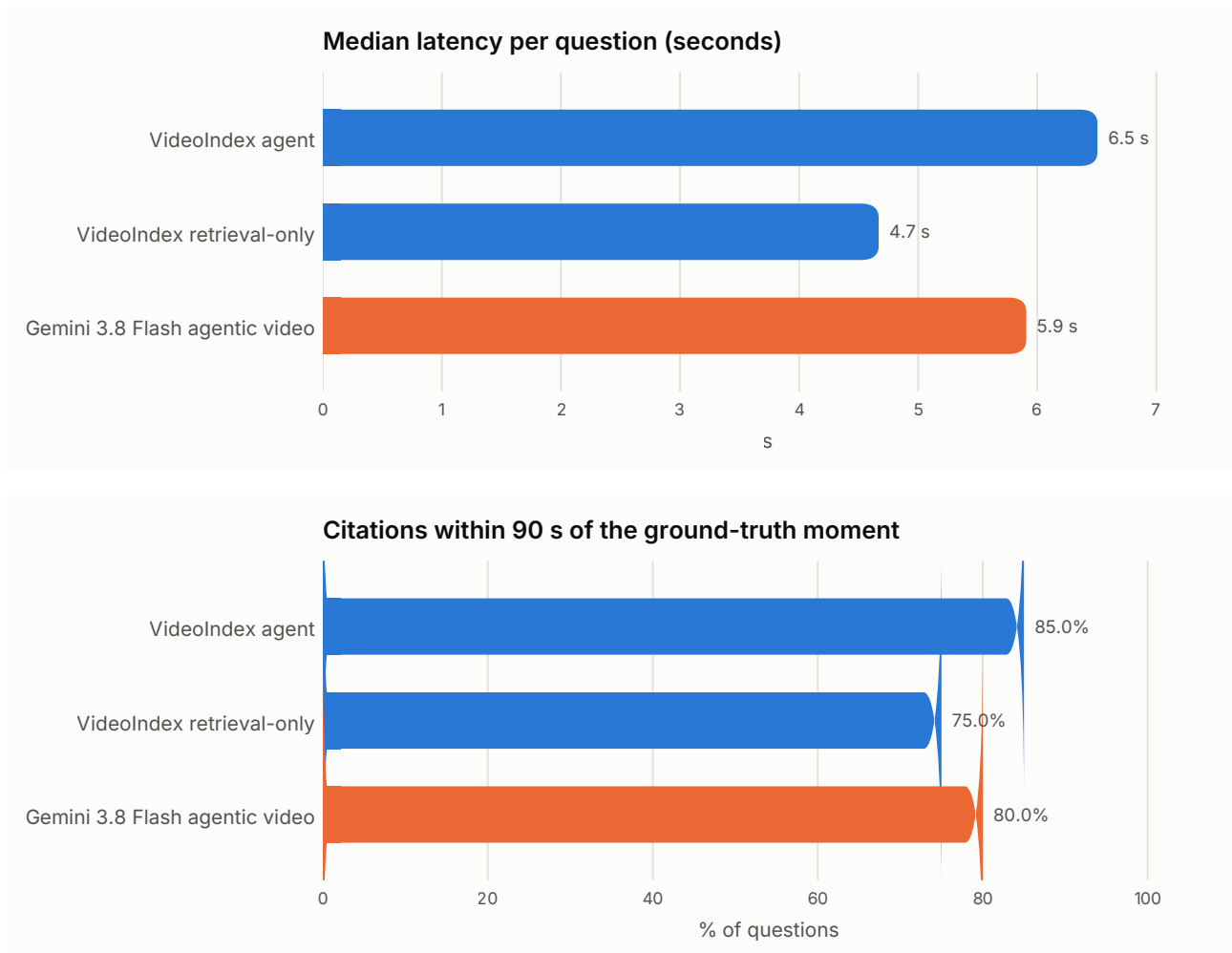
Like-for-like run: same questions, same files

`scripts/gemini_video_eval.py` asked **Gemini 3.8 Flash** (`gemini-3.8-flash`) with `processing: "agentic"` the QA dev set's questions over the same video files (each file uploaded once through the Files API; one `interactions.create` call per question; the prompt asks for a one- or two-sentence answer with `[HH:MM:SS]` citations). Answers were scored with the QA rules above: substring match against the accepted strings, and a citation counts as in time when a timestamp in the answer lies within 90 s of the anchor. Gemini cost is computed from the returned usage object at list price: prompt text and the media the agent loaded (`total_tool_use_tokens`) at \$0.75 per million, cached tokens at \$0.075 per million, output and thinking tokens at \$3.75 per million. The run was stopped after 20 questions by decision: this is a development set, and the extensive comparison is planned on LVBench and the other public benchmarks in M4. The VideoIndex rows in this section are recomputed over the same 20 questions so the comparison is like for like; the full-set VideoIndex numbers are in the previous section.

System (20 questions)	accuracy	cited	citation within 90 s	cost / question	tokens / question	median latency	p90 latency
VideoIndex agent (Claude Sonnet 5 over the index)	100.0%	100.0 %	85.0%	\$0.049	14,821	6.5 s	9.1 s
VideoIndex retrieval-only	75.0%	80.0 %	75.0%	\$0.017	5,252	4.7 s	5.6 s
Gemini 3.8 Flash agentic video	95.0%	100.0 %	80.0%	\$0.047	15,267	5.9 s	14.3 s

Gemini's agent made 1.8 `processing_call` steps per question on average, loading 3,063 tokens of frames and transcript, and spent 11,794 thinking tokens per question; thinking is the largest cost component at output prices. VideoIndex's token count is Claude's context across all agent calls (index text, never pixels unless `view` is called).





Reading the comparison

- **Different models, different work.** Gemini answers from the raw video with its own model; VideoIndex answers from a prebuilt index with Claude Sonnet 5. The comparison is of two products on one task, not of two LLMs.
- **Indexing cost is not in the per-question figures.** VideoIndex spends compute once per video (about 2 to 12 minutes of machine time per hour of video on this host depending on the build and the slide density); those costs amortise over every question asked of the video. Gemini's per-question cost is the whole cost.
- **Citations are typed differently.** VideoIndex citations are machine-checkable spans tied to index rows; Gemini's are timestamps the model writes in prose, parsed from the text. Both are scored by the same 90 s rule.
- **The scoring is the same, generous substring rule for both,** so short accepted strings (a single word) can match inside a longer wrong answer for either system.

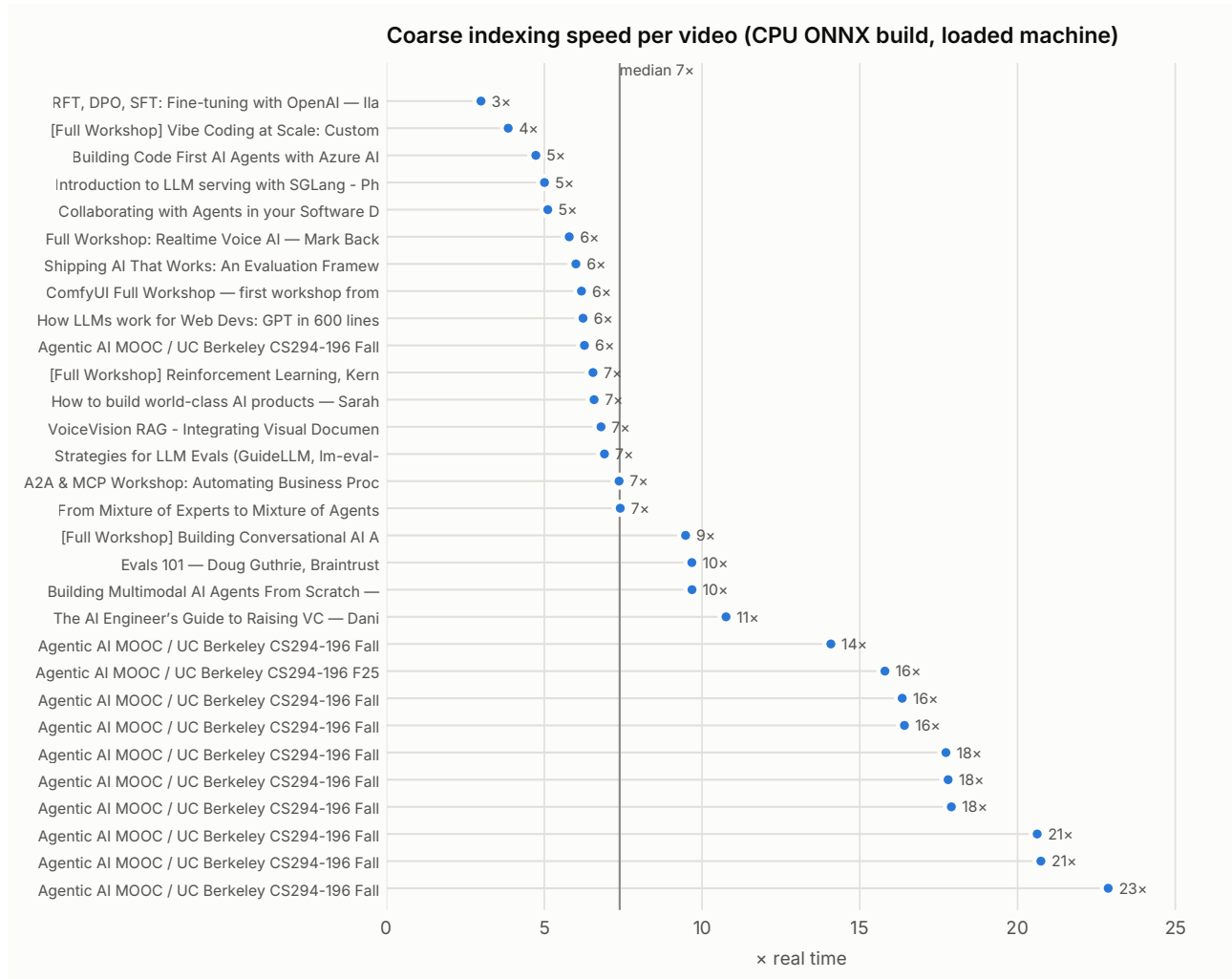
Gemini misses

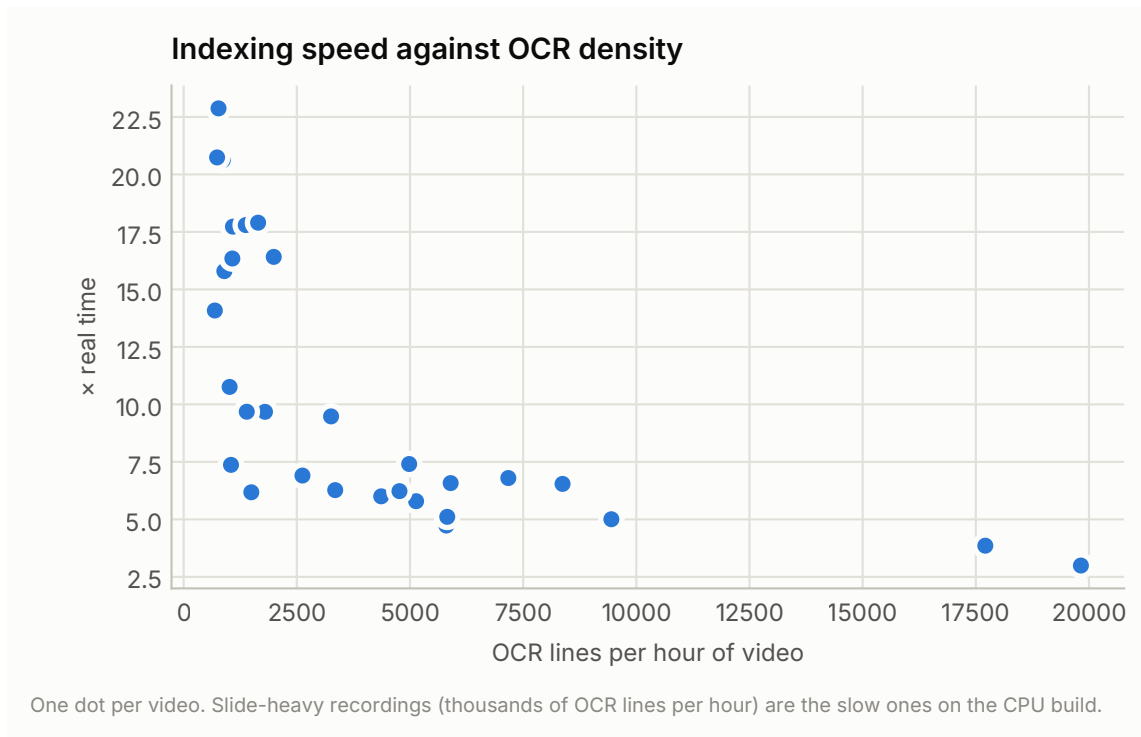
Question	Accepted	Answer given (truncated)
qa13: In Sida Wang's lecture, which benchmark is said to often need hundreds of thousands of tok	terminal bench, terminal-bench, terminal	In Sida Wang's lecture, agent evaluations such as SWE-bench (including its various versions) and T-Bench are noted as often requirin

Results: indexing throughput

The dataset run

The 30 videos (36:34:35) were indexed in one `vi_index` process with the `coarse_only` policy in 17,856 s (5.0 h), 7× real time overall, with zero failures. The machine was not idle: release builds, test suites and a second index were running at times, so the per-video numbers are an upper bound on the cost of the CPU build.





The spread is explained by OCR density. Camera-heavy talks index at 20× real time or better; slide decks that change every few seconds produce thousands of OCR lines per hour, and on the CPU build RapidOCR's detector and recogniser dominate. Whisper ran on the GPU throughout at about 29× real time for the speech it was given.

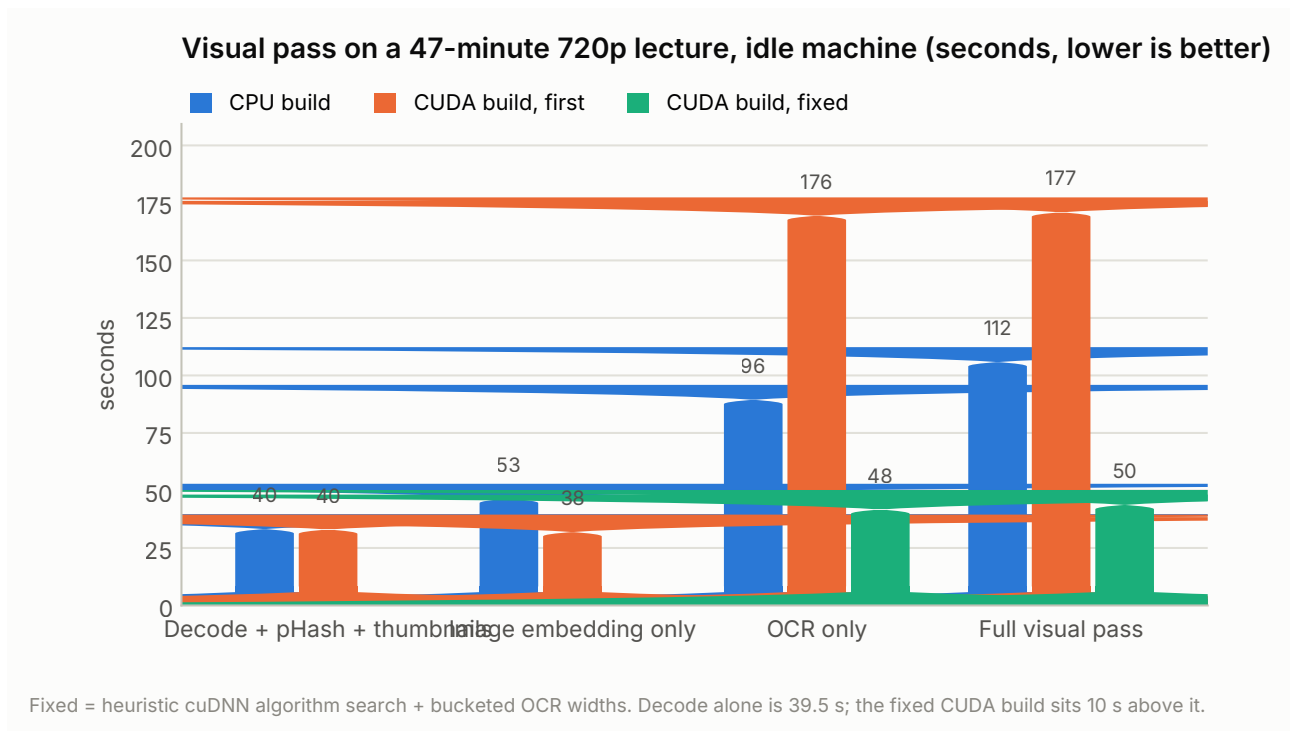
Per-video timings (CPU ONNX build, loaded machine)

Video	Duration	Wall	x real time	asr	ocr	image_embed	shot_boundary	text_embed
[Full Workshop] Reinforcement Learning, Kernels, Reasoning,	2:42:27	1490 s	7×	86 3	226 63	220	65	24225
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / LLM Agen	1:58:21	400 s	18×	62 3	214 8	154	87	3839
Building Code First AI Agents with Azure AI Agent Service —	1:54:05	1447 s	5×	76 2	1103 4	759	106	12289
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Agentic	1:48:50	398 s	16×	58 3	361 0	77	39	5225
RFT, DPO, SFT: Fine-tuning with OpenAI — Ilan Bigio, OpenAI	1:46:14	2130 s	3×	65 2	350 98	368	77	36160
How to build world-class AI products — Sarah Sachs (AI lead	1:43:45	946 s	7×	55 6	102 00	553	66	11214
How LLMs work for Web Devs: GPT in 600 lines of Vanilla JS -	1:41:33	978 s	6×	56 7	807 0	415	83	9065
Shipping AI That Works: An Evaluation Framework for PMs – Am	1:26:16	862 s	6×	45 7	627 5	733	76	7109

Video	Duration	Wall	× real time	asr	ocr	image_embed	shot_boundary	text_embed
VoiceVision RAG - Integrating Visual Document Intelligence w	1:23:51	740 s	7×	503	10024	250	65	10898
A2A & MCP Workshop: Automating Business Processes with LLMs	1:23:13	678 s	7×	417	1451	801	61	2236
[Full Workshop] Vibe Coding at Scale: Customizing AI Assista	1:20:38	1255 s	4×	541	23800	370	66	24694
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Evolutio	1:19:31	209 s	23×	428	1021	80	36	2147
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Post-Tra	1:17:28	225 s	21×	414	1117	13	14	2219
Full Workshop: Realtime Voice AI — Mark Backman, Daily	1:09:40	722 s	6×	383	5963	559	47	6641
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Training	1:04:37	275 s	14×	366	742	121	23	1673
Collaborating with Agents in your Software Dev Workflow - Jo	1:04:06	752 s	5×	368	6221	486	44	6854
[Full Workshop] Building Conversational AI Agents - Thor Sch	1:01:42	391 s	9×	347	3349	245	41	3971
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Autonomo	1:01:42	590 s	6×	326	3443	469	103	4319
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / AI Agent	1:01:15	206 s	18×	327	1401	79	26	2272
Agentic AI MOOC / UC Berkeley CS294-196 F25 / Multi-Agent Sy	0:58:58	224 s	16×	309	883	166	50	1719
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Multi-Ag	0:54:49	159 s	21×	320	675	42	18	1484
From Mixture of Experts to Mixture of Agents with Super Fast	0:53:15	431 s	7×	301	4421	119	38	4955
ComfyUI Full Workshop — first workshop from ComfyAnonymous h	0:51:25	499 s	6×	297	1281	595	31	1810
Evals 101 — Doug Guthrie, Braintrust	0:48:31	301 s	10×	258	1453	183	51	1920
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Practica	0:46:54	172 s	16×	252	841	168	41	1509

Video	Duration	Wall	× real time	asr	ocr	image_embed	shot_boundary	text_embed
Agentic AI MOOC / UC Berkeley CS294-196 Fall 2025 / Predicta	0:44:04	148 s	18×	258	1208	37	17	1875
Introduction to LLM serving with SGLang - Philip Kiely and Y	0:43:42	524 s	5×	254	6882	114	35	7330
Building Multimodal AI Agents From Scratch — Apoorva Joshi,	0:36:58	229 s	10×	207	859	133	45	1227
The AI Engineer's Guide to Raising VC — Dani Grant (Jam), Ch	0:34:16	191 s	11×	197	580	47	6	922
Strategies for LLM Evals (GuideLLM, lm-eval-harness, OpenAI	0:32:28	282 s	7×	177	1420	149	37	1740

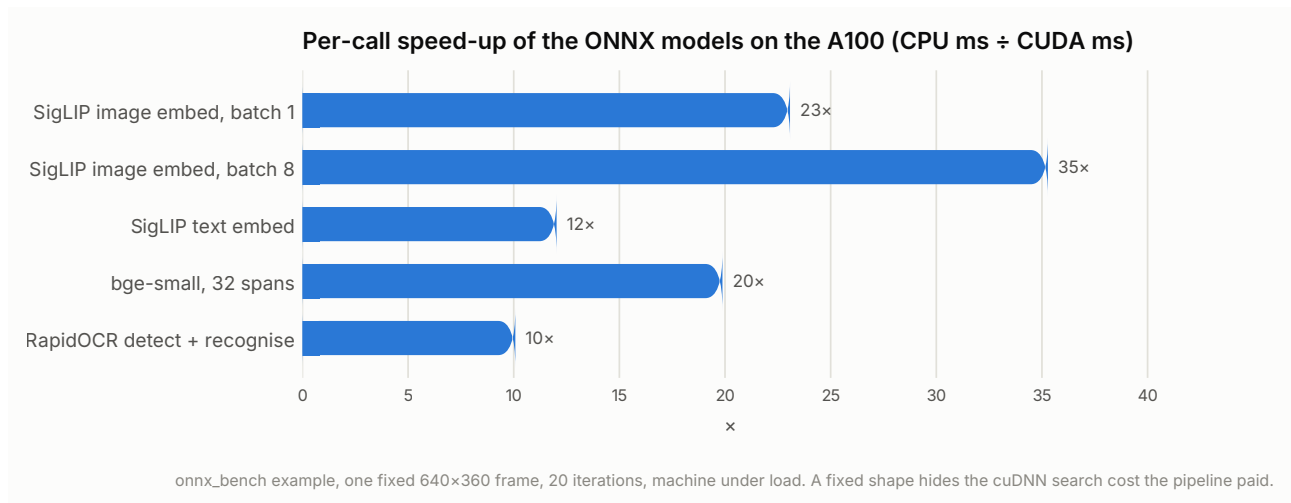
CPU against GPU



Policy (47-minute 720p VP9 lecture, idle machine)	CPU build	CUDA build, first	CUDA build, fixed
Decode + pHash + thumbnails	39.7 s	39.5 s	—
Image embedding only	52.9 s	38.4 s	—
OCR only	95.9 s	175.9 s	48.2 s
Full visual pass	112.3 s	177.4 s	50.2 s

Decode alone (sampling, pHash, thumbnails) takes 39.5 s for this 47-minute video, 71× real time, on either build. The first CUDA build lost time in OCR: cuDNN's default exhaustive convolution-algorithm search re-benchmarks every convolution for each new input shape, and the recogniser's padded batch width changed on almost every call. With the heuristic search and recogniser widths rounded up to multiples of

32, OCR-only drops from 175.9 s to 48.2 s and the full visual pass to 50.2 s, 10 s above the decode floor. The per-call benchmark below, taken on a single fixed frame, could not see this: it shows the GPU 10 to 35× faster per call, which is true once shapes repeat.



Call	CPU (6 threads)	CUDA	Speed-up
SigLIP image embed, batch 1	270 ms	11.7 ms	23×
SigLIP image embed, batch 8	1623 ms	46.0 ms	35×
SigLIP text embed	83 ms	6.9 ms	12×
bge-small, 32 spans	165 ms	8.3 ms	20×
RapidOCR detect + recognise	262 ms	26.0 ms	10×

Reproducing

```
# retrieval
python3 scripts/devset_eval.py /data/videoindex/indexes/dataset.vidx --config config/gcp-a100.toml --json /data/videoindex/eval/retrieval2.json
# question answering, agent and baseline
python3 scripts/devset_qa_eval.py /data/videoindex/indexes/dataset.vidx --config config/gcp-a100.toml --json /data/videoindex/eval/qa-agent2.json
python3 scripts/devset_qa_eval.py /data/videoindex/indexes/dataset.vidx --config config/gcp-a100.toml --policy retrieval-only --json /data/videoindex/eval/qa-retrieval-only.json
# Gemini agentic video on the same questions and files (needs GEMINI_API_KEY)
python3 scripts/gemini_video_eval.py --model gemini-3.8-flash --mode agentic --json /data/videoindex/eval/qa-gemini-agentic.json
# this report
python3 scripts/report/benchmark_report.py
```